

Neuro-Symbolic Safety Nets: Fusing Optimized Deep Learning with Interpretable Logic for Trustworthy Diagnosis

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Abstract—The “black box” nature of Deep Learning hinders its adoption in high-stakes medical domains. We propose a Hybrid Neuro-Symbolic Framework that fuses an optimized Multi-Layer Perceptron (MLP) with an interpretable symbolic expert layer. Using the CDC Diabetes Dataset, we implement a “Total Defense” strategy involving robust outlier detection (Isolation Forest) and rigorous Hyperparameter Optimization (HPO) via Optuna and Enhanced ALSHADE. Our system achieves a peak neural ROC-AUC of 0.82. While integrating the symbolic layer reduces discrimination (ROC-AUC 0.79), it acts as a critical safety audit, identifying 20.36% of false negatives missed by data-driven models. This work demonstrates a viable path toward trustworthy, human-in-the-loop AI that prioritizes safety alongside accuracy.

Index Terms—Neuro-Symbolic AI, Hyperparameter Optimization, Explainable AI, Medical Diagnosis, Trustworthy AI, Autoencoders, Evolutionary Computing

I. INTRODUCTION

Deep Learning offers high predictive accuracy but lacks the interpretability required for clinical trust. Purely data-driven models are often brittle, failing on edge cases without explanation. To bridge this “Trust Gap,” we present a “Hybrid Neuro-Symbolic Diagnostic System” combining the statistical power of neural networks with transparent, logic-driven reasoning.

Our key contributions are:

- 1) **Robust Preprocessing:** Evaluation of Isolation Forest [1] against standard Z-score pruning.
- 2) **Hybrid Representation:** Comparison of raw features, PCA, and Autoencoder embeddings [2].
- 3) **Extensive Optimization:** Tuning of six optimizers (including Lion [3]) using Random Search, Optuna, and Evolutionary Algorithms (e.g., Enhanced ALSHADE [4]).
- 4) **Neuro-Symbolic Layer:** Integration of medical rules, fuzzy logic, and Bayesian updates to audit neural predictions.
- 5) **Safety Analysis:** Demonstration that hybrid systems expose limitations of pure Deep Learning on difficult patient profiles.

II. RELATED WORK

While architectures like TabNet exist, MLPs remain effective for tabular data when properly regularized. Neuro-symbolic approaches often embed logic into training; our method uses a symbolic layer as an independent post-hoc auditor. We systematically compare HPO methods (Bayesian vs. Evolutionary) to optimize this neural baseline, contrasting with standard grid search approaches.

III. METHODOLOGY

A. Dataset Preprocessing

We utilize the “CDC Diabetes Health Indicators Dataset”. To address the 1:7 class imbalance (Fig. 1), we implemented a “Total Defense” strategy: 1. “Isolation Forest”: Removes anomalies based on feature splits rather than statistical tails (Fig. 2). 2. “Random Oversampling”: Balances the minority class. 3. “Weighted Loss”: BCEWithLogitsLoss prioritizes sensitivity. 4. “Dynamic Thresholding”: Maximizes F1-Score on validation data.

TABLE I
DATASET CHARACTERISTICS (CDC DIABETES HEALTH INDICATORS)

Characteristic	Value
Total Samples	253,680
Total Features	21
Class 0 (Non-Diabetic)	218,334 (86.1%)
Class 1 (Diabetic)	35,346 (13.9%)
Class Ratio	1 : 6.2
Missing Values	0
Feature Types	Numerical, Binary

Figures 3, 4, and 5 highlight the complex, non-linear feature relationships.

B. Representation Learning

We investigate three input formats: Raw features, “PCA” (linear projection), and “Autoencoder (AE)” embeddings (non-linear compression). PCA visualizations (Figs. ??-7) reveal significant class overlap. UMAP (Fig. 8) confirms Isolation Forest preserves manifold structure better than Z-score.

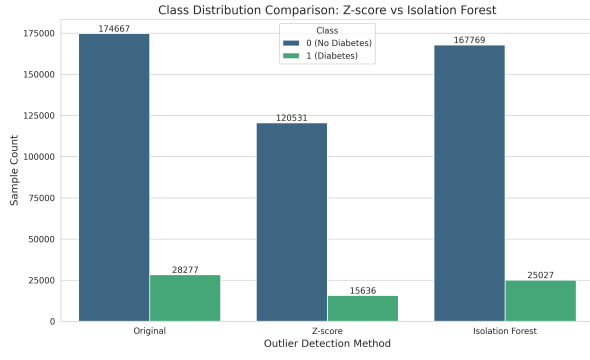


Fig. 1. Class Distributions: Original vs. Z-score vs. Isolation Forest.

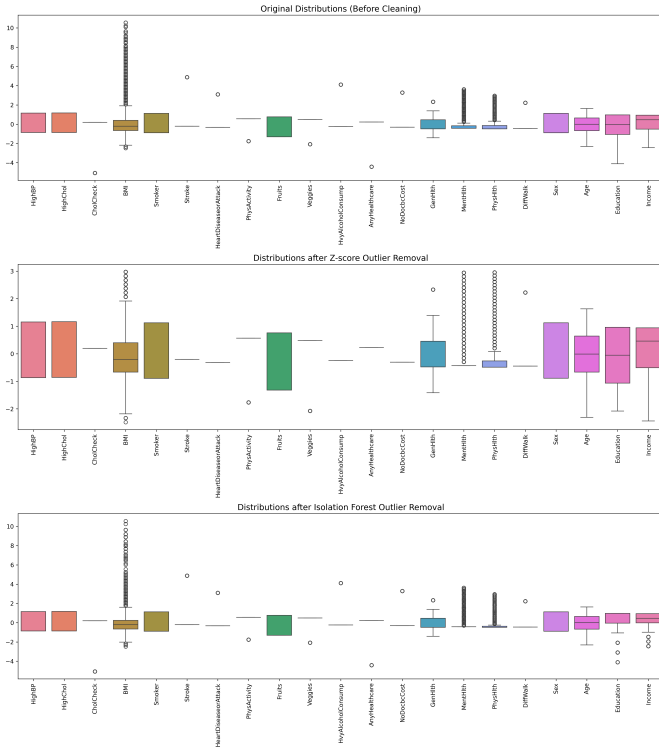


Fig. 2. Feature distributions: Isolation Forest vs. Z-score.

The AE minimizes reconstruction loss $\mathcal{L}_{AE} = ||\mathbf{x} - g_{\phi}(f_{\theta}(\mathbf{x}))||^2 + \lambda ||\theta||^2$ to capture latent features (Figs. 9-10).

C. Baseline Neural Network HPO

We employ a fixed MLP (Fig. 11) with two dense layers (32, 16 neurons, ReLU) and a sigmoid output. We optimized hyperparameters (Learning Rate, Batch Size, Decay, Dropout) using: **Random Search**: Stochastic baseline. **Optuna**: Bayesian Optimization (TPE). **Evolutionary Algorithms**: GA, DE, PSO, and Enhanced ALSHADE [4].

D. Neuro-Symbolic Interpretability Layer

To audit the MLP, we implement a parallel expert layer:
 1. **Rule-Based**: Explicit heuristics (e.g., $BMI \geq 30 \wedge Age \geq 45 \rightarrow Risk$).
 2. **Fuzzy Logic**: Models continuous

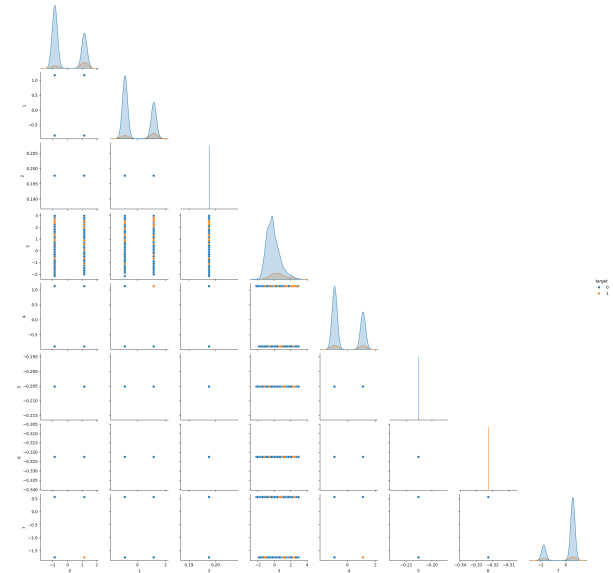


Fig. 3. Pairplot of selected features.



Fig. 4. Feature distributions.

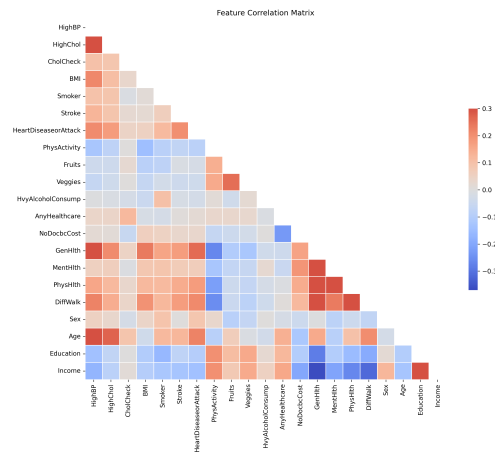


Fig. 5. Correlation heatmap.

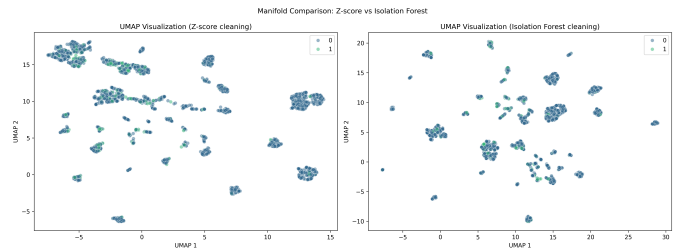


Fig. 8. UMAP: Z-score vs. Isolation Forest.

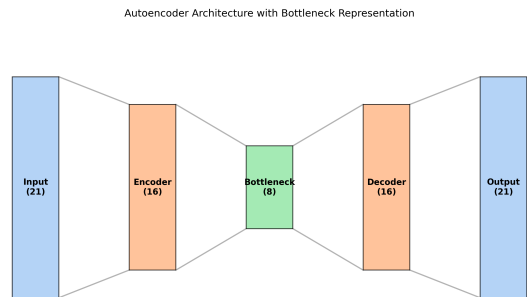


Fig. 9. Autoencoder Architecture.

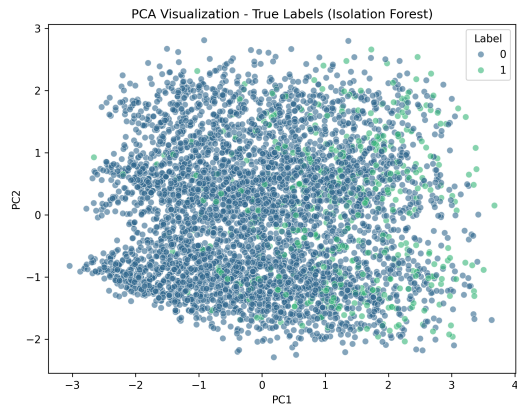


Fig. 6. PCA Visualization (Isolation Forest).

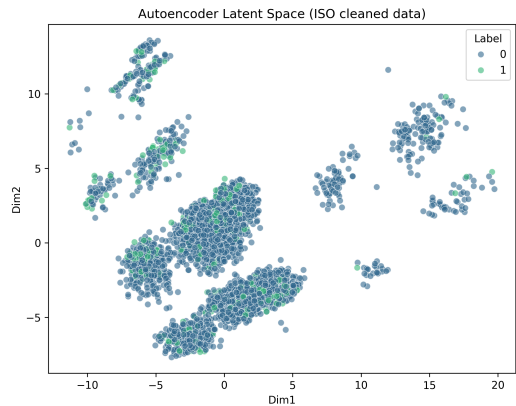


Fig. 10. AE Latent Space (Isolation Forest).

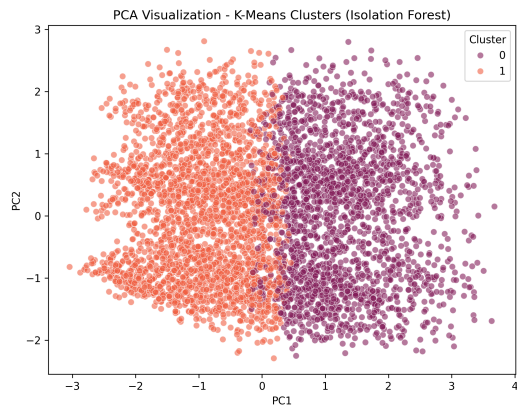


Fig. 7. PCA K-Means Clusters.

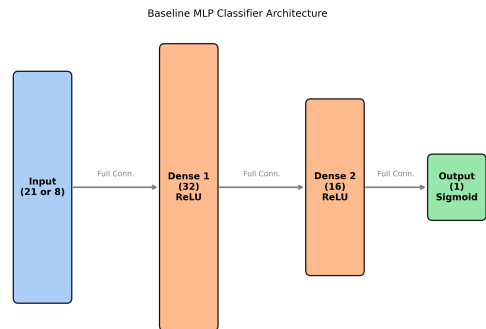


Fig. 11. Baseline MLP Architecture.

variables (Age, BMI) into "Low/Medium/High" risk scores.

3. **Bayesian Update:** A Gaussian Naive Bayes (GNB) classifier provides an independent probabilistic baseline.

Decision Fusion: The final probability p_{hybrid} is the average of MLP, Fuzzy, and GNB scores, overridden by explicit rules for high-confidence safety guarantees (Fig. 12).

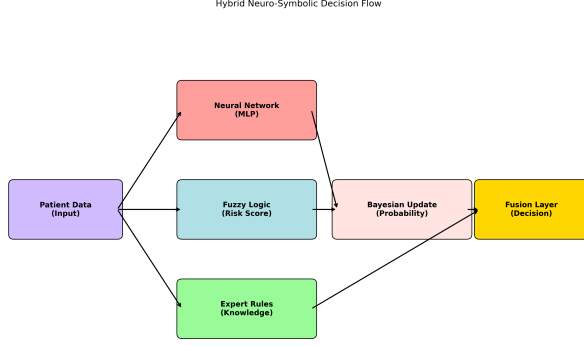


Fig. 12. Hybrid Neuro-Symbolic Decision Flow.

IV. EXPERIMENTS RESULTS

A. Performance HPO Convergence

We evaluated models using Accuracy, Precision, Recall, F1, and ROC-AUC. Figure 13 shows Optuna converging fastest ($ROC \approx 0.83$). Table IV confirms Optuna and Enhanced ALSHADE as top performers.

TABLE II
PERFORMANCE COMPARISON WITH BASELINE MODELS

Model	Accuracy	F1-Score	AUC-ROC	MCC
Logistic Regression	0.7929	0.3600	0.7674	0.2426
Ridge Classifier	0.8605	0.0344	0.8134	0.0713
k-Nearest Neighbors	0.8475	0.2673	0.7172	0.2081
Decision Tree	0.7913	0.2993	0.5939	0.1778
Gaussian NB	0.7574	0.3929	0.7754	0.2780
Bernoulli NB	0.7521	0.4021	0.7721	0.2910
AdaBoost	0.8625	0.2391	0.8162	0.2297
Gradient Boosting	0.8642	0.2385	0.8225	0.2362
Extra Trees	0.8506	0.2331	0.7657	0.1890
Bagging Classifier	0.8445	0.2595	0.7389	0.1961
Random Forest	0.8556	0.2255	0.7873	0.1970
SVM (RBF)	0.8619	0.0548	0.6645	0.1073
Linear SVC	0.7993	0.3280	0.7491	0.2113
SGD Classifier	0.8596	0.1949	0.8003	0.1902
Proposed (Hybrid)	0.5959	0.3606	0.7103	-

TABLE III
HYPERPARAMETER OPTIMIZATION (HPO) STRATEGY ANALYSIS (BEST CONFIGURATIONS)

Feature	Optimizer	HPO Method	Rec	F1	ROC	Rank
Raw	SGD	Bayesian Opt.	0.1200	0.1961	0.8193	2
PCA	Lion	Bayesian Opt.	0.1166	0.1936	0.8212	1
AE	Adagrad	Genetic Algo.	0.0000	0.0000	0.7552	3

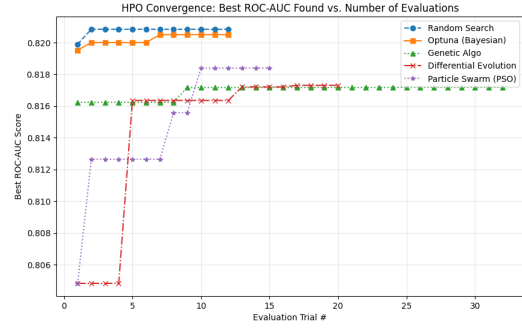


Fig. 13. HPO Convergence: Optuna vs. GA vs. Random.

Hyperparameter distributions (Figs. ??, 14) show that the "Total Defense" pipeline stabilizes performance across the search space.

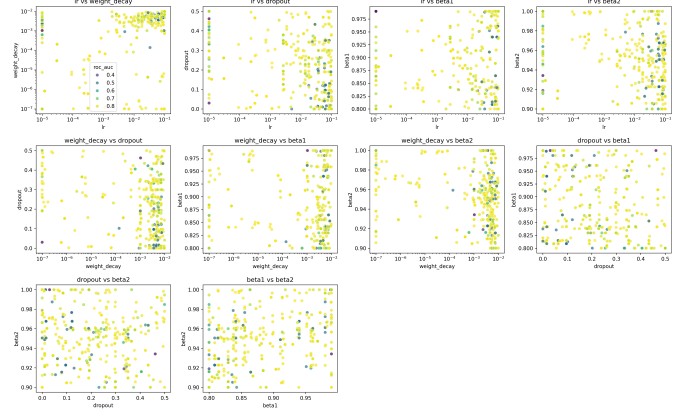


Fig. 14. Hyperparameter Distributions (Isolation Forest).

TABLE IV
HPO COMPARISON

Method	Best ROC-AUC	Time (s)
Random Search	0.8286	176.5
Bayesian (Optuna)	0.8322	2143.5
Genetic Algorithm	0.8252	1140.5
Enhanced ALSHADE	0.8264	1984.2

B. Calibration Analysis

The optimized MLP shows good calibration (Fig. 15) and appropriately shifts risk for borderline BMI/Age cases (Figs. 16, ??). Feature importance (Fig. 17) confirms reliance on valid medical factors like General Health and BMI.

TABLE V
REPRESENTATION LEARNING PERFORMANCE

Representation	Acc	F1	ROC	Notes
Raw Features	0.8629	0.1961	0.8193	Best: SGD / Bayesian Opt.
PCA Features	0.8647	0.1936	0.8212	Best: Lion / Bayesian Opt.
AE Embeddings	0.8607	0.0000	0.7552	Best: Adagrad / Genetic Algo.

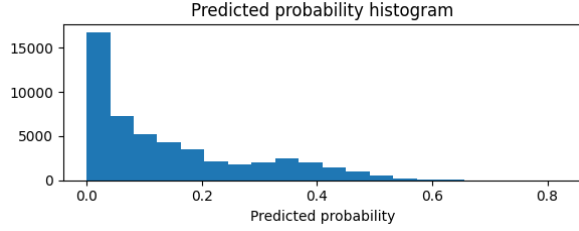


Fig. 15. Reliability Diagram.

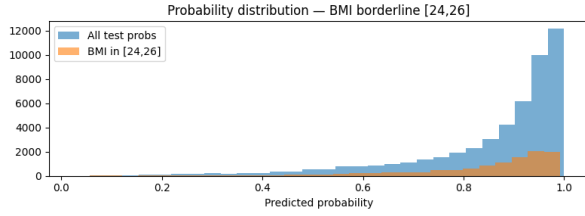


Fig. 16. Borderline BMI analysis.

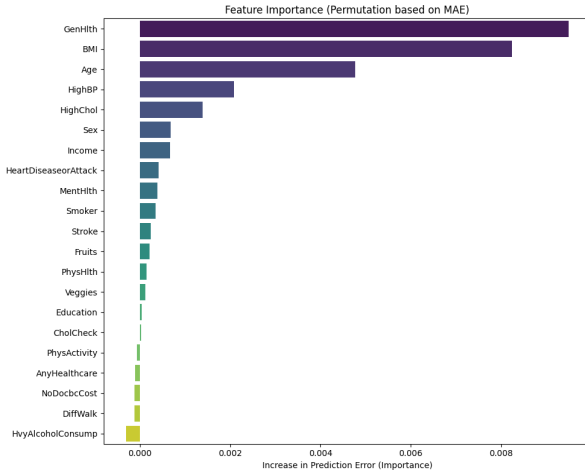


Fig. 17. Permutation Feature Importance.

C. Neuro-Symbolic Validation

Table VI and Table VII highlight the system's safety. While the MLP achieved higher raw AUC (0.82 vs 0.79), the Hybrid system recovered ****2,245 False Negatives**** (20.36% reduction).

The ablation study (Figs. ??-19) confirms that the fuzzy-Bayesian layer provides crucial robustness.

TABLE VI
HARD FALSE NEGATIVES ANALYSIS

ID	True	MLP	Fuzzy	Bayes
46974	1	0.0005	0.0476	0.0000
5789	1	0.0013	0.0435	0.0000

TABLE VII
CONCRETE AUDITING OF DIAGNOSTIC FAILURES (FN REDUCTION)

Model	TP	FP	TN	FN	FN Rate
MLP (PCA+Lion)	4250	7085	36582	2819	39.88%
Hybrid System	4824	10353	33314	2245	31.76%
Improvement	20.36% reduction in False Negatives				

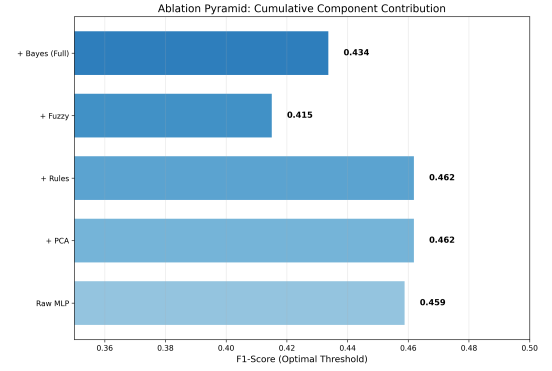


Fig. 18. Ablation Pyramid.

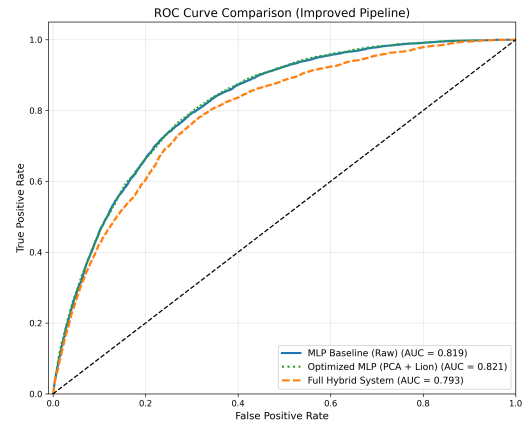


Fig. 19. ROC Curve: MLP vs. Hybrid.

V. CONCLUSION

We optimized a deep learning baseline to a peak **ROC-AUC of 0.8212** (PCA + Lion). However, by integrating a Neuro-Symbolic safety layer, we traded slight statistical discrimination (0.79 AUC) for a **20% reduction in missed diagnoses**. This "Total Defense" approach demonstrates that hybrid AI can function as an effective safety net, validating the potential for trustworthy, human-in-the-loop medical systems.

APPENDIX: REPRODUCIBILITY

Rules: 1) ObeseOld $\rightarrow$ High Risk. 2) HighBPChol $\rightarrow$ High Risk. 3) Sedentary $\rightarrow$ High Risk. 4) Frail $\rightarrow$ High Risk. 5) Healthy $\rightarrow$ Low Risk. **Fuzzy Sets:** BMI Low/Med/High; Age Young/Mid/Old (Triangular functions). **HPO Budgets:** 15 trials (Random/Optuna); 32 evals (Evolutionary).

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SUPPLEMENTARY MATERIAL

TABLE VIII
HYPERPARAMETER OPTIMIZATION (HPO) STRATEGY ANALYSIS FOR RAW INPUT (ISO)

Feature	Optimizer	HPO Method	Acc	Prec	Rec	F1	ROC	Epochs	Run-Time	p-value	F-Rank
Raw Input	RMSProp	Random Search	0.7880	0.3487	0.6011	0.4413	0.8036	15	211.9s	n.s.	21
		Bayesian Opt.	0.8014	0.3661	0.5816	0.4493	0.8094	66	4926.3s	↓ 0.001	14
		Genetic Algo.	0.7821	0.3456	0.6312	0.4466	0.8096	36	1329.5s	↓ 0.001	13
		ALSHADE	0.8025	0.3690	0.5881	0.4534	0.8136	26	1749.3s	n.s.	6
	Adam	Random Search	0.8020	0.3702	0.5999	0.4578	0.8150	15	189.5s	↓ 0.001	4
		Bayesian Opt.	0.7901	0.3534	0.6108	0.4478	0.8083	64	2332.0s	↓ 0.001	16
		Genetic Algo.	0.7850	0.3494	0.6302	0.4495	0.8108	36	1654.2s	↓ 0.001	11
		ALSHADE	0.8064	0.3726	0.5692	0.4504	0.8111	22	2355.1s	↓ 0.001	10
	SGD	Random Search	0.7941	0.3599	0.6141	0.4538	0.8149	15	228.6s	↓ 0.001	5
		Bayesian Opt.	0.7994	0.3675	0.6104	0.4588	0.8193	71	2565.1s	↓ 0.001	1
		Genetic Algo.	0.7998	0.3665	0.5990	0.4547	0.8157	32	1140.5s	↓ 0.001	3
		ALSHADE	0.7900	0.3504	0.5941	0.4408	0.8065	32	1651.1s	↓ 0.001	18
	Adagrad	Random Search	0.7998	0.3602	0.5629	0.4393	0.8026	15	187.5s	↓ 0.001	22
		Bayesian Opt.	0.7802	0.3422	0.6261	0.4426	0.8045	28	2155.9s	↓ 0.001	19
		Genetic Algo.	0.8030	0.3685	0.5804	0.4508	0.8129	32	1859.8s	↓ 0.001	8
		ALSHADE	0.7924	0.3544	0.5968	0.4447	0.8078	50	1984.2s	↓ 0.001	17
	Adadelata	Random Search	0.8022	0.3677	0.5835	0.4511	0.8100	15	197.8s	↓ 0.001	12
		Bayesian Opt.	0.7945	0.3589	0.6040	0.4503	0.8090	23	1449.4s	↓ 0.001	15
		Genetic Algo.	0.8067	0.3753	0.5824	0.4565	0.8171	32	1591.7s	↓ 0.001	2
		ALSHADE	0.7830	0.3409	0.5970	0.4340	0.8011	60	2427.9s	↓ 0.001	23
	Lion	Random Search	0.7873	0.3527	0.6306	0.4524	0.8119	15	176.5s	↓ 0.001	9
		Bayesian Opt.	0.7950	0.3604	0.6089	0.4528	0.8135	74	1980.7s	↓ 0.001	7
		Genetic Algo.	0.6264	0.2405	0.7790	0.3675	0.7175	36	1199.0s	↓ 0.001	24
		ALSHADE	0.7888	0.3508	0.6067	0.4446	0.8043	28	1265.0s	↓ 0.001	20

TABLE IX
HYPERPARAMETER OPTIMIZATION (HPO) STRATEGY ANALYSIS FOR PCA (ISO)

Feature	Optimizer	HPO Method	Acc	Prec	Rec	F1	ROC	Epochs	Run-Time	p-value	F-Rank
PCA	RMSProp	Random Search	0.8132	0.3845	0.5670	0.4583	0.8202	15	175.4s	n.s.	8
		Bayesian Opt.	0.8004	0.3690	0.6096	0.4597	0.8207	62	2143.5s	↓ 0.001	3
		Genetic Algo.	0.8013	0.3697	0.6043	0.4588	0.8168	36	1198.2s	↓ 0.001	19
		ALSHADE	0.7925	0.3609	0.6347	0.4602	0.8198	26	1726.2s	↓ 0.001	14
	Adam	Random Search	0.8133	0.3845	0.5664	0.4581	0.8204	15	192.1s	↓ 0.001	5
		Bayesian Opt.	0.8073	0.3765	0.5840	0.4579	0.8188	71	2335.8s	↓ 0.001	17
		Genetic Algo.	0.7953	0.3630	0.6214	0.4583	0.8189	36	1053.0s	↓ 0.001	16
		ALSHADE	0.7994	0.3680	0.6124	0.4597	0.8201	58	1419.6s	↓ 0.001	12
	SGD	Random Search	0.8066	0.3763	0.5903	0.4596	0.8202	15	155.0s	↓ 0.001	7
		Bayesian Opt.	0.7747	0.3368	0.6366	0.4405	0.8037	77	2111.3s	↓ 0.001	23
		Genetic Algo.	0.8061	0.3756	0.5915	0.4594	0.8206	32	853.1s	↓ 0.001	4
		ALSHADE	0.7074	0.2600	0.5960	0.3621	0.7288	37	1395.3s	↓ 0.001	24
	Adagrad	Random Search	0.8126	0.3839	0.5700	0.4588	0.8201	15	171.1s	↓ 0.001	13
		Bayesian Opt.	0.7962	0.3645	0.6223	0.4597	0.8196	60	2093.5s	↓ 0.001	15
		Genetic Algo.	0.7973	0.3656	0.6188	0.4596	0.8202	32	1388.6s	↓ 0.001	11
		ALSHADE	0.7862	0.3548	0.6526	0.4597	0.8202	45	3700.5s	↓ 0.05	6
	Adadelata	Random Search	0.7958	0.3627	0.6154	0.4564	0.8174	25	557.3s	↓ 0.001	18
		Bayesian Opt.	0.7944	0.3595	0.6089	0.4521	0.8125	46	1585.8s	↓ 0.001	21
		Genetic Algo.	0.8112	0.3824	0.5770	0.4599	0.8211	32	2705.8s	↓ 0.001	2
		ALSHADE	0.7915	0.3605	0.6415	0.4616	0.8202	13	3026.5s	↓ 0.001	10
	Lion	Random Search	0.7935	0.3613	0.6281	0.4587	0.8202	29	514.2s	↓ 0.001	9
		Bayesian Opt.	0.8048	0.3749	0.6012	0.4619	0.8212	47	1388.2s	↓ 0.001	1
		Genetic Algo.	0.7794	0.3448	0.6482	0.4502	0.8081	36	2528.1s	↓ 0.001	22
		ALSHADE	0.8016	0.3693	0.5992	0.4570	0.8157	49	1979.9s	↓ 0.05	20

TABLE X
HYPERPARAMETER OPTIMIZATION (HPO) STRATEGY ANALYSIS FOR AUTOENCODER (ISO)

Feature	Optimizer	HPO Method	Acc	Prec	Rec	F1	ROC	Epochs	Run-Time	p-value	F-Rank
Autoencoder	RMSProp	Random Search	0.7871	0.3038	0.4087	0.3485	0.6793	14	510.2s	n.s.	19
		Bayesian Opt.	0.7609	0.3013	0.5427	0.3875	0.7473	71	1538.9s	↓ 0.001	8
		Genetic Algo.	0.7225	0.2789	0.6258	0.3859	0.7489	36	2906.0s	↓ 0.001	7
		ALSHADE	0.5618	0.2072	0.7591	0.3256	0.6906	49	0.0s	↓ 0.001	17
	Adam	Random Search	0.7847	0.3014	0.4139	0.3488	0.7079	15	565.4s	↓ 0.001	16
		Bayesian Opt.	0.7643	0.3018	0.5265	0.3837	0.7423	71	1649.7s	↓ 0.001	12
		Genetic Algo.	0.7154	0.2799	0.6629	0.3936	0.7537	36	2415.2s	↓ 0.001	3
		ALSHADE	0.7533	0.2945	0.5526	0.3843	0.7540	34	3086.2s	↓ 0.001	2
	SGD	Random Search	0.7706	0.2915	0.4518	0.3544	0.7223	25	470.4s	↓ 0.001	14
		Bayesian Opt.	0.7939	0.3123	0.3988	0.3503	0.7170	71	1320.0s	↓ 0.001	15
		Genetic Algo.	0.7141	0.2769	0.6529	0.3889	0.7530	36	2571.5s	↓ 0.001	4
		ALSHADE	0.7415	0.2816	0.5511	0.3727	0.7442	44	0.0s	↓ 0.001	11
	Adagrad	Random Search	0.7310	0.2805	0.5947	0.3812	0.7516	25	901.1s	↓ 0.001	5
		Bayesian Opt.	0.1393	0.1393	1.0000	0.2446	0.4555	46	1795.0s	↓ 0.001	23
		Genetic Algo.	0.7490	0.3007	0.6046	0.4016	0.7552	32	3316.5s	↓ 0.001	1
		ALSHADE	0.1393	0.1393	1.0000	0.2446	0.4239	30	4257.3s	↓ 0.001	24
	Adadelta	Random Search	0.7345	0.2806	0.5791	0.3781	0.7458	25	1119.4s	↓ 0.001	10
		Bayesian Opt.	0.1393	0.1393	1.0000	0.2446	0.4574	46	2484.8s	↓ 0.001	22
		Genetic Algo.	0.7479	0.2922	0.5690	0.3861	0.7499	32	3151.6s	↓ 0.001	6
		ALSHADE	0.7477	0.2854	0.5391	0.3732	0.7462	58	3659.1s	↓ 0.001	9
	Lion	Random Search	0.7872	0.3042	0.4098	0.3492	0.6617	15	1106.6s	↓ 0.001	20
		Bayesian Opt.	0.7731	0.3083	0.5054	0.3830	0.7284	46	2373.2s	↓ 0.001	13
		Genetic Algo.	0.6889	0.2525	0.6287	0.3602	0.6864	36	3789.4s	↓ 0.001	18
		ALSHADE	0.1393	0.1393	1.0000	0.2446	0.5000	60	3813.3s	↓ 0.001	21